

Assignment_CloudFormation_2

AWS CloudFormation: Rollback

1. What is rollback in CloudFormation?

Rollback in CloudFormation means returning the stack to a safe previous state when a stack operation fails. If CloudFormation is creating, updating, or deleting resources and something goes wrong, it tries to undo the failed changes automatically.

For example, if a stack is creating an EC2 instance, VPC, and S3 bucket, but the EC2 instance fails because of an invalid AMI ID or permission issue, CloudFormation does not leave the stack half-created. Instead, it starts rollback and removes the resources that were already created during that failed operation.

Rollback is important because it protects the infrastructure from being left in an incomplete or broken condition. It also makes troubleshooting easier because the user can check the stack events, find the exact failed resource, fix the template or permission issue, and then try again.

During an update, rollback works in a similar way. If the update fails, CloudFormation attempts to restore the stack to the last working configuration. This is useful in real projects because it reduces manual cleanup and helps keep infrastructure stable.

In simple words, rollback is CloudFormation's safety mechanism. It helps recover from failed deployments by undoing unsuccessful changes and keeping the stack controlled.